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1  # Note 09: Decision Tree Algorithm
2  # =====
3
4  # Import Libraries
5  import pandas as pd
6  from sklearn.model_selection import train_test_split
7  from sklearn.tree import DecisionTreeClassifier
8  from sklearn.metrics import accuracy_score, confusion_matrix
9  from sklearn.tree import plot_tree
10 import matplotlib.pyplot as plt
11
12 # Load dataset
13 iris = pd.read_csv("Iris.csv")
14
15 # Features and Target
16 X = iris.drop(columns=["Species", "Id"])
17 y = iris["Species"]
18
19 # Split dataset
20 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=
42)
21
22 # Create Decision Tree model
23 model = DecisionTreeClassifier(random_state=42, criterion="gini", max_depth=None,
max_leaf_nodes=None)
24
25 # Train model
26 model.fit(X_train, y_train)
27
28 # Predict
29 y_pred = model.predict(X_test)
30
31 # Header
32 print("\n=====")
33 print("Evaluation Report - Decision Tree Classifier")
34 print("=====\n")
35
36 # Print Results
37 print("Accuracy as a Percentage :", round((accuracy_score(y_test, y_pred)*100), 2),
"\b%")
38
39 # Header
40 print("\n=====")
41 print("Confusion Matrix")
42 print("=====\n")
43 print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred), "\n")
44
45 # Predict a New Flower
46 new_flower = pd.DataFrame([[5.1, 3.5, 1.4, 0.2]], columns=["SepalLengthCm",
"SepalWidthCm", "PetalLengthCm", "PetalWidthCm"])
47 prediction = model.predict(new_flower)
48 print("Expected Flower : Iris-setosa")
49 print("Predicted Flower : ", prediction[0])
50
51 # Plot Decision Tree
52 plt.figure()
53 plot_tree(model, feature_names=X.columns.tolist(), class_names=model.classes_, filled=
True, rounded=True,)
54 plt.title("Trained Decision Tree Structure")
55 plt.show()
56
57 # Print Header
58 print("\n=====")
59 print("Important Features")
60 print("=====\n")
61
62 # Find Important Features
63 for feature, importance in zip(X_train.columns, model.feature_importances_):
64     print(feature, ":", importance)

```